◇ Problem Statement

> Create a **Document Editor** where user can add:

- Text
- Images

📌 And system should be **scalable**, so that in future we can add:

- Video
- New line
- Tab space
- Other document elements

👉 **Follow SOLID principles** while designing.

◇ Approach to Solve LLD Problems

1️⃣ Top-Down Approach

- Pehle **main object** banao
- Phir **small objects**

2️⃣ Bottom-Up Approach ☑

- Pehle **small objects**
- Phir **main object**

📌 **Generally used by ~90% developers**

👉 **Is problem me Bottom-Up approach use kiya gaya hai**

🧠 *Reason*: Chhote-chhote components clear hote hain → system scalable hota hai.

🟤 Initial Design – BAD DESIGN

◇ DocumentEditor (Bad Design)

🧠 Idea

- Text aur Image dono ko **string** ki tarah store kar rahe hain
- Kyunki hume nahi pata user kis order me add karega

🖉 Text Diagram – Bad Design

```
+--------------------------------+
|        DocumentEditor          |
|--------------------------------|
| vector<string> elements        |
|--------------------------------|
| addText(string text)           |
| addImage(string path)          |
| renderDocument()               |
| saveToFile()                   |
+--------------------------------+
```

❌ Problems in Bad Design

1. ❌ **Breaks OCP**
    - New feature (video, newline) add karne ke liye → existing code modify karna padega
2. ❌ **Breaks SRP**
    - Rendering
    - Element handling
    - Saving sab ek hi class me
3. ❌ **Not Implemented**
    - LIP ❌
    - DIP ❌
    - ISP ❌

😵 *Basically ek "God Class" ban gayi hai*

◇ BETTER DESIGN – Proper Explanation

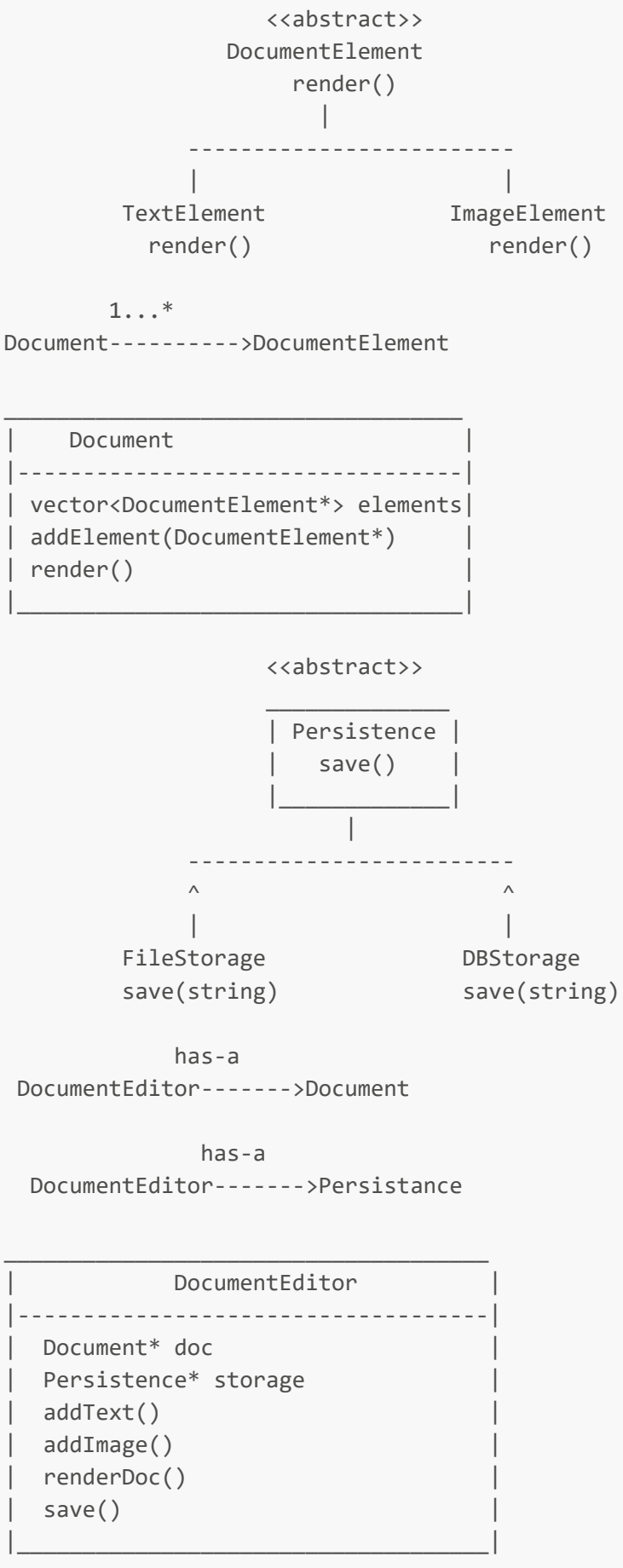
📌 Goal of Better Design

Bad design ke problems ko solve karna:

- `if-else` logic hataana
- Hard-coding kam karna
- Abstraction introduce karna

Lekin abhi **SRP fully follow nahi hota**.

🖉 TEXT UML DIAGRAM – BETTER DESIGN (Document Editor)


<<abstract>>
DocumentElement
render()
TextElement
render()
ImageElement
render()
1...*
Document
DocumentElement
Document
vector<DocumentElement*> elements
addElement(DocumentElement*)
render()
<<abstract>>
Persistence
save()
FileStorage
save(string)
DBStorage
save(string)
has-a
DocumentEditor
Document
has-a
DocumentEditor
Persistance
DocumentEditor
Document* doc
Persistence* storage
addText()
addImage()
renderDoc()
save()
Client
DocumentEditor

🧩 1. DocumentElement (Abstraction)

```
<<abstract>>
DocumentElement
    render()
```

☑ Kya karta hai?

- Ye ek **base abstraction** hai
- Har document ka element **render()** karega

🧠 Why needed?

- Agar kal **VideoElement / TableElement** add karna ho → existing code modify nahi karna padega ✔ **OCP follow**

🧩 2. TextElement & ImageElement

```
TextElement      ImageElement
 render()          render()
```

☑ Kya karta hai?

- Har element **apna render logic khud define** karta hai
- Document ko ye jaanne ki zarurat nahi:
    - kaunsa text hai
    - kaunsa image hai

🧠 Benefit

- `if (type == image)` jaisa logic remove ho gaya ✔ **Polymorphism used**

🧩 3. Document (Better design ka weak point)

```
Document
vector<DocumentElement*> elements
addElement()
render()
```

❌ Problem yahin se start hoti hai

Document:

- elements **store bhi** karta hai
- elements ko **render bhi** karta hai

👉 **Two responsibilities**

- Data holding
- Rendering logic

❌ **SRP violation**

🧩 4. Persistence Abstraction

```
<<abstract>>
Persistence
 save()
```

☑ Kya karta hai?

- Saving ka contract define karta hai
- DocumentEditor ko ye nahi pata:
  - file me save ho raha hai
  - DB me save ho raha hai

✔ **DIP follow**

🧩 5. FileStorage & DBStorage

```
FileStorage      DBStorage
 save()            save()
```

🧠 Benefit

- Storage change karne pe editor ka code change nahi hota ✔ **OCP + DIP**

🧩 6. DocumentEditor (Another weak point)

```
DocumentEditor
- addText()
- addImage()
- renderDoc()
- save()
```

❌ Problem

DocumentEditor:

- elements create karta hai
- rendering trigger karta hai
- persistence call karta hai

👉 **Coordinator + Creator + Controller ❌ Multiple responsibilities**

🎯 BETTER DESIGN SUMMARY

✔ Bad design se kaafi better

✔ Abstraction + Polymorphism used

❌ SRP fully satisfied nahi

❌ Document & Editor overloaded

📌 **Interview line**

> Better design removes bad practices but may still have mixed responsibilities.

◇ OPTIMAL DESIGN – Proper Explanation

📌 Goal of Optimal Design

> **Har class = ek responsibility**

Aur:

- Easily extendable
- Easily testable
- Production-grade LLD

🖊 TEXT UML DIAGRAM – OPTIMAL DESIGN

```
                <<abstract>>
                DocumentElement
                     |
              -------------
              |           |
         TextElement  ImageElement
           render()     render()

        1...*
Document--------->DocumentElement
```

```
    <<model>>
_____________________________________
|    Document                       |
|-----------------------------------|
| vector<DocumentElement*> elements|
| addElement(DocumentElement*)     |
| getElement()                      |
|___________________________________|


DocumentRenderer-------->Document
_____________________
|     DocumentRenderer  |
|-----------------------|
| render(Document)      |
|_______________________|


                   <<abstract>>
                    Persistence
                   save(string data)
                       |
              -------------------------
              |                       |
         FileStorage             DBStorage
         save(string)            save(string)



DocumentEditor----------->Document

DocumentEditor----------->Persistance
_____________________________________
|            DocumentEditor          |
|------------------------------------|
|  Document* doc                     |
|  Persistence* storage              |
|  addText()                         |
|  addImage()                        |
|____________________________________|

Client  --------------------->  Persistnce

Client  --------------------> Document

Client  --------------------->  DocumentEditor

Client  --------------------->  Documentation
```

🧩 1. DocumentElement hierarchy (same, but clean)

```
DocumentElement
   |
TextElement   ImageElement
```

☑ Same benefit

- Polymorphism
- OCP
- No `if-else`

🧩 2. Document (ONLY data holder now)

```
Document
- vector<DocumentElement*>
- addElement()
- getElement()
```

☑ Improvement

- **Rendering hata diya**
- Ab sirf:
    - structure maintain karta hai

✔ **SRP satisfied**

🧩 3. DocumentRenderer (New class – BIG WIN)

```
DocumentRenderer
 render(Document)
```

🧠 Kya solve hua?

- Rendering logic **Document se bahar**
- Agar kal:
    - HTML render
    - PDF render
    - Markdown render

👉 New renderer add karo, document untouched

✔ **SRP + OCP**

🧩 4. Persistence (same as before, but cleaner)

```
Persistence
 save(data)
```

✔ Benefit

- Editor sirf abstraction jaanta hai
- Storage change → editor safe

✔ **DIP**

🧩 5. DocumentEditor (Now PURE coordinator)

```
DocumentEditor
- uses Document
- uses Persistence
```

☑ Ab kya karta hai?

- Client input leta hai
- Correct object ko delegate karta hai

❌ Rendering logic nahi

❌ Storage logic nahi

✔ **Single Responsibility**

🧩 6. Client Dependency (Correct flow)

```
Client -> DocumentEditor
```

Client directly:

- storage
- document
- renderer

pe depend nahi karta

✔ **Loose coupling**

🎯 OPTIMAL DESIGN SUMMARY

✔ Strict SRP

✔ Fully SOLID compliant

✔ Easily extensible

✔ Interview-ready architecture

🔥 Short Summary: Better vs Optimal Design (Core Difference)

🟠 Better Design

> Better design me **abstraction hoti hai**, lekin classes ko **abhi bhi doosri layers ki knowledge hoti hai**.

Tumhara exact point:

- `Document` ko pata hai `render()` ka concept (jabki rendering ka kaam `DocumentElement` / renderer ka hona chahiye)
- `DocumentEditor` ko pata hai:
  - save ho raha hai
  - `Persistence` ka `save()` method exist karta hai

👉 Matlab:

- **Editor ko storage ke behavior ka idea hai**
- **Document ko rendering ka idea hai**

❌ **Knowledge leak ho rahi hai**

❌ **Responsibilities thodi mix hain**

📌 Isliye:

> Better design **working hai**, but **loosely coupled nahi hai**

🟠 Optimal Design

> Optimal design me **class sirf apna kaam jaanti hai**, baaki sab ka kaam **delegate** kar deti hai.

Same scenario but clean way:

- `Document`:

    - sirf elements store karta hai
    - render ka concept bhi nahi jaanta

- `DocumentEditor`:
    - sirf coordinator hai
    - usse nahi pata:
        - file me save ho raha hai
        - DB me save ho raha hai
        - kaise render ho raha hai

👉 Matlab:

- **No extra knowledge**
- **No responsibility leakage**
- **True abstraction**

✔ **Loose coupling**

✔ **Pure SRP**

✔ **Future-proof design**

🎯 One-Line Final Difference (Interview Gold)

> **Better design reduces problems, but optimal design removes unnecessary knowledge between classes.**

Agar ek aur line bolni ho:

> **In better design, classes still know “how”; in optimal design, they only know “what”.**

Tumhari thinking bilkul sahi direction me ja rahi hai bhaiya 🧠 Ye wahi observation hai jo **LLD interview me candidate ko stand-out** karwati hai.